

# Disproof of Conjecture 00000003991: the functional pigeonhole principle has short cutting-plane refutations

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October 2026

## 1 The conjecture and its reading

**Statement (English text of the problem).** *Definition: The functional pigeonhole principle  $PHP\_n$  is the formalization of the absence of an injection from  $n+1$  to  $n$ . Conjecture: The cutting plane proof length lower bound of  $PHP\_n$  is  $2^{n/8}$ , and the exponential constant  $1/8$  cannot be improved under cutting plane rules with bounded coefficients.* The Chinese text says the same: the lower bound for the proof length of  $PHP\_n$  in the cutting-plane system is  $2^{n/8}$ , and the constant  $1/8$  cannot be improved under cutting-plane rules with bounded coefficients.

**Reading.** The conjecture is a conjunction. Its first clause says that every cutting-plane (CP) refutation of  $PHP\_n$  has length at least  $2^{n/8}$ . We read this in its weakest form, “for all sufficiently large  $n$ ”. The “for all  $n$ ” version is stronger, so it fails too. The second clause (“ $1/8$  cannot be improved”) presupposes the first. We refute the first clause, and with it the conjunction. We show:

1. For every  $n \geq 1$ ,  $PHP\_n$  has a CP refutation with at most  $8n^3$  lines. Every coefficient and every constant term in it has absolute value at most  $2n$ .
2. Hence, for every  $\varepsilon > 0$  and  $c > 0$ , there is no lower bound  $c \cdot 2^{\varepsilon n}$ , even for all large  $n$ . In particular there is no lower bound  $2^{n/8}$ . This also holds when only refutations whose coefficients are bounded by  $2n$  are counted, i.e. CP with polynomially bounded coefficients.

**Readings that are not addressed.** (a) The reading in which *both* clauses concern CP whose coefficients are bounded by an absolute constant independent of  $n$ . Our coefficients grow like  $2n$ , so we do not address this reading. (b) Clause 2 taken on its own, as “no lower bound  $2^{cn}$  with  $c > 1/8$  holds under bounded coefficients”. Our result is consistent with this statement, so we do not claim that it is false. (c) “Length” always means the number of lines, as in the definition quoted below. Bit size is not formalized in Lean (but see Remark 5).

## 2 Definitions (as in the Lean file)

**Cutting planes.** Dantchev, Galesi, Ghani and Martin [2] (arXiv:2102.07622) attribute the definition to Cook, Coullard and Turán [1]. In their words: “The Cutting Planes (CP) proof system is equipped with boolean axioms and two inference rules”: the Boolean axioms  $x \geq 0$  and  $-x \geq -1$ ; Linear Combination, from  $\mathbf{ax} \geq c$  and  $\mathbf{bx} \geq d$  infer  $\alpha\mathbf{ax} + \beta\mathbf{bx} \geq \alpha c + \beta d$ ; and Rounding, from  $\alpha\mathbf{ax} \geq b$  infer  $\mathbf{ax} \geq \lceil b/\alpha \rceil$ . Moreover, “A CP refutation of some unsatisfiable set of integer linear inequalities is a derivation of  $0 \geq 1$  by the aforementioned inference rules from the inequalities in  $\mathcal{F}$ ”, and “The length of a CP refutation is the number of inequalities in the sequence.” Beame et al. [3] (arXiv:1710.03219) state the rules as follows: “From inequalities  $c_i x \geq d_i, c_j x \geq d_j$ , deduce any non-negative linear

combination with integer coefficients”, and “From an inequality  $c_i x \geq d_i$ , if  $t \in \mathbb{Z}$  divides all entries in  $c_i$  then deduce  $(c_i/t)x \geq \lceil d_i/t \rceil$ ”.

In Lean (**C3991.Line**, **Step**, **IsDerivation**, **IsRefutation**) a line is a pair  $(a, b)$  with  $a : V \rightarrow \mathbb{Z}$  and  $b \in \mathbb{Z}$ . It stands for  $\sum_v a_v x_v \geq b$ . A **Step** from the axiom set **Ax** and the earlier lines is one of the following: a member of **Ax**; a Boolean axiom  $x_v \geq 0$  or  $-x_v \geq -1$ ;  $\alpha A + \beta B$  for earlier lines  $A, B$  and integers  $\alpha, \beta > 0$ ; or  $(a/\alpha, \lceil b/\alpha \rceil)$  for an earlier line  $(a, b)$  and an integer  $\alpha > 0$  that divides every  $a_v$ . The ceiling is taken in  $\mathbb{Q}$ . A derivation is a list of lines in which every line is a **Step** from the lines strictly before it. A refutation is a derivation whose last line is  $0 \geq 1$ . Its length is the number of lines. Each of our rules is an instance of the rules quoted from [3]: we use positive rather than non-negative multipliers. So a short refutation in our system is also a short refutation in theirs. As in [3], the constant term  $b$  of the rounding rule may be any integer: this is the standard Cutting Planes rounding rule with integer right-hand side, as stated in [3], and it is the rule our construction uses (it rounds the negative constant  $-(2m-1)$ ). The rule table in [2] writes  $b \in \mathbb{Z}^+$ ; we do not rely on any inference about that table, and our disproof is for standard CP as just stated. **Bdd B L** means that all coefficients and the constant term of  $L$  have absolute value at most  $B$ .

**The functional pigeonhole principle.** [2] defines  $\text{PHP}_n^m$  as the inequalities  $\sum_{j=1}^n P_{i,j} \geq 1$  (“every pigeon goes into some hole”) and  $P_{i,k} + P_{j,k} \leq 1$  for  $i \neq j$  (“at most one pigeon enters any given hole”). The functional version adds that no pigeon is in two holes. In Lean, **Var n** is  $\text{Fin}(n+1) \times \text{Fin } n$ , and **FPHP n** consists of the following axioms:

- **pigeonAx i**:  $\sum_j x_{i,j} \geq 1$ ;
- **holeAx i i' j**:  $-x_{i,j} - x_{i',j} \geq -1$  for  $i \neq i'$ ;
- **funAx i j j'**:  $-x_{i,j} - x_{i,j'} \geq -1$  for  $j \neq j'$ .

**Faithfulness** (**fphp\_sat\_iff**). A 0/1 assignment satisfies all of **FPHP n** (**Sat**:  $b \leq \sum_v a_v x_v$ ) if and only if it is the indicator of the graph of an injective map  $\text{Fin}(n+1) \rightarrow \text{Fin } n$ . By **fphp\_unsat**, no such assignment exists. So **FPHP n** is exactly the formalization of the absence of an injection from  $n+1$  to  $n$ .

### 3 The short refutation

Fix  $n \geq 1$  and a hole  $j$ . Let  $T_{j,m}$  be the line  $-\sum_{i < m} x_{i,j} \geq -1$  (**T j m**). The pigeons are  $0, \dots, n$ .

**Lemma 1** (**hole\_step**, **C\_zero**, **C\_succ**, **round\_C**). *Let  $2 \leq m \leq n$ . From  $T_{j,m}$  one derives  $T_{j,m+1}$  with  $2m+1 \leq 2n+1$  further lines.*

*Proof.* Write down the hole axioms  $h_k = \text{holeAx } k \ m \ j$  for  $k = 0, \dots, m-1$ , and form  $C_0 = (m-1)T_{j,m} + h_0$  and  $C_k = C_{k-1} + h_k$ . Then  $C_{m-1} = (m-1)T_{j,m} + \sum_{k < m} h_k$  is the line  $-m \sum_{i \leq m} x_{i,j} \geq -(2m-1)$  (**C\_last\_coeff**). Every coefficient is divisible by  $m$ . Rounding by  $m$  gives  $-\sum_{i \leq m} x_{i,j} \geq \lceil -(2m-1)/m \rceil = -1$ , which is  $T_{j,m+1}$ .  $\square$

The line  $T_{j,2}$  is the hole axiom **holeAx 0 1 j** (**T\_two**). Iterating the lemma gives  $T_{j,n+1}$ , i.e.  $-\sum_i x_{i,j} \geq -1$ , with at most  $1+n(2n+1)$  lines per hole (**hole\_full**). Doing this for all  $n$  holes takes at most  $n(1+n(2n+1))$  lines (**all\_holes**). Next, the pigeon axioms are summed into  $G_n : \sum_{i,j} x_{i,j} \geq n+1$  with  $2n+1$  lines (**pigeons**). Finally, adding the  $n$  lines  $T_{j,n+1}$  to  $G_n$  gives  $H_n : 0 \geq (n+1) - n = 1$  with  $n$  further lines (**holes\_sum**, **H\_n**). The derivation is cut off after its first  $0 \geq 1$  line (**refutation\_of\_mem**). The total length is at most  $2n^3 + n^2 + 4n + 1 \leq 8n^3$ . The functionality axioms are not used. They are only part of the axiom set. The largest coefficients occur in  $C_k$ : they are  $-m$  and  $-(k+1)$ , with constant term  $-(m+k)$ . All of these have absolute value at most  $2n$ .

**Theorem 2** (`fphp_short_refutation`). *For every  $n \geq 1$  there is a list  $l$  with `IsRefutation` (FPHP  $n$ )  $l$ ,  $|l| \leq 8n^3$ , and `Bdd` ( $2n$ )  $L$  for every line  $L$  of  $l$ .*

The exact length is  $n^3 + 2n^2 + n + 1$ . A Python check rebuilt the refutation and verified it line by line for  $n \leq 8$ . Lean proves only the bound  $8n^3$ .

**Theorem 3** (`eventually_poly_lt`, `fphp_subexponential`, `no_exponential_lower_bound`). *For all  $\varepsilon, c > 0$ , for all sufficiently large  $n$ , there is a refutation of FPHP  $n$  with coefficients bounded by  $2n$  and fewer than  $c \cdot 2^{\varepsilon n}$  lines. Hence no  $c > 0$  satisfies “for all large  $n$ , every refutation with coefficients bounded by  $2n$  has length  $\geq c \cdot 2^{\varepsilon n}$ ”.*

*Proof.* Mathlib’s `isLittleO_pow_exp_pos_mul_atTop` gives  $x^3 = o(e^{\varepsilon \log 2 \cdot x})$ , and  $e^{\varepsilon \log 2 \cdot n} = 2^{\varepsilon n}$ . Concretely,  $8n^3 < 2^{n/8}$  for all  $n \geq 209$ .  $\square$

**Theorem 4** (`conjecture_3991_false`). *Both of the following statements are false: “for all large  $n$ , every CP refutation  $l$  of FPHP  $n$  satisfies  $2^{n/8} \leq |l|$ ”; and the same statement restricted to refutations with coefficients bounded by  $2n$ .*

**Remark 5.** This remark is not formalized. Each line has  $n(n+1)$  coefficients and one constant term, each of absolute value at most  $2n$ . So the bit size of the refutation is also polynomial in  $n$ . The result is the classical fact, attributed in [2] to [1], that PHP has “polynomial size proofs in CP”. Quoted from [2]: “appealing to the polynomial size proofs in CP of the  $\text{PHP}_n^m$  ([CCT87])”.

## 4 Lean verification

File `lean/Conjecture3991/Basic.lean` (namespace `C3991`) imports only Mathlib. It contains no `sorry`, no `native_decide` and no added axioms. `#print axioms` for `conjecture_3991_false`, `no_exponential_lower_bound`, `fphp_subexponential`, `fphp_short_refutation`, `fphp_sat_iff` and `fphp_unsat` reports only `propext`, `Classical.choice` and `Quot.sound`.

## References

- [1] W. Cook, C. R. Coullard, Gy. Turán, On the complexity of cutting-plane proofs, *Discrete Applied Mathematics* (1987), [https://doi.org/10.1016/0166-218x\(87\)90039-4](https://doi.org/10.1016/0166-218x(87)90039-4). (Cited through [2].)
- [2] S. Dantchev, N. Galesi, A. Ghani, B. Martin, Depth lower bounds in Stabbing Planes for combinatorial principles, <https://arxiv.org/abs/2102.07622>.
- [3] P. Beame, N. Fleming, R. Impagliazzo, A. Kolokolova, D. Pankratov, T. Pitassi, R. Robere, Stabbing Planes, <https://arxiv.org/abs/1710.03219>.